Example 3.10

Rani earns ₹ 200 on the first day and spends some amount in the evening. She earns ₹ 300 on the second day and spends double the amount as she spents on the first day. She earns ₹ 400 on the third day and spends 4 times the amount as she spents on the first day. Can you give an algebraic expression of the total amount with her, at the end of the third day.

Solution

The amount earned on the first day is ₹ 200.

Let the amount spent on the first day be ₹ x. Amount with her at the end of the first day is (200 – x).

Amount earned on the second day is ₹ 300 and the amount spent on the second day is ₹ $2x$. The amount left on the second day is 300 – $2x$.

Similarly, the net amount that she would have on the third day is 400 – $4x$. Therefore, the total amount that she would have at the end of three days is (200 – x) + (300 – $2x$) + (400 – $4x$).

That is, $200 + 300 + 400 + (-1 - 2 - 4)x$

$= 900 + (-7)x$

Thus the required algebraic expression is 900 – $7x$.

Activity

Here is a magic number trick that you can perform using Algebra.

Choose any number of your choice	If it is 2	If it is 3	If it is 4	General term 'x'
Add 2 to this number	2 + 2 = 4	3 + 2 = 5	4 + 2 = 6	$x + 2$
Multiply the sum by 5	4 × 5 = 20	5 × 5 = 25	6 × 5 = 30	$5(x + 2) = 5x + 10$
Add 10 to the product	20 + 10=30	25 + 10 = 35	30+10= 40	$5x + 10 + 10 = 5x + 20 = 5(x+4)$
Divide the sum by 5	$\frac{30}{5} = 6$	$\frac{35}{5} = 7$	$\frac{40}{5} = 8$	$\frac{5(x+4)}{5} = x + 4$
Subtract the original number	6 – 2 = 4	7 – 3 = 4	8 – 4 = 4	$x + 4 - x = 4$
The final result is	4	4	4	4

Everyone will get the same result 4 irrespective of the number they think. Play this number game with your friends and surprise them. We generalized the pattern in the final column using algebraic expressions. Observe the same and create your own number game and verify whether it works for any number by finding the general term.

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